

Lab 34: Pandas Grouping and Aggregation

Aim

To group data by a categorical column and compute aggregate statistics for each group.

Algorithm

1. Import the Pandas library.
2. Create a DataFrame containing class and marks data.
3. Group the data based on the class column.
4. Calculate the mean marks for each class.
5. Perform multiple aggregation operations such as mean, maximum, and minimum.
6. Display the results.

Program

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```
import pandas as pd
```

```
# Create DataFrame
```

```
df = pd.DataFrame({  
    "class": ["I", "II", "I", "II", "I", "II"],  
    "marks": [78, 85, 92, 66, 74, 88]  
})
```

```
# Calculate mean marks for each class
```

```
print("Mean marks per class:\n")  
print(df.groupby("class")["marks"].mean())
```

```
# Perform multiple aggregation operations
```

```
print("\nMultiple aggregates:\n")
```

```
print(df.groupby("class")["marks"].agg(  
    ["mean", "max", "min"]  
))
```

Sample Output

Mean marks per class:

class

I 81.333333

II 79.666667

Name: marks, dtype: float64

Multiple aggregates:

mean max min

class

I 81.333333 92 74

II 79.666667 88 66

Result

Thus, **grouping and multiple aggregation operations** were successfully performed using Pandas to calculate statistical values for each group.